Deterministic Optimization

Convex Conic Programming

Introduction

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Convex Cones, Order, and Linear Conic Programming

Nonlinear Optimization Modeling

Learning Objectives for this module

- Explore generalizations of linear programming to nonlinear programming through convex cones and generalized inequalities
- Understand second-order cone and semidefinite cone and associated conic optimization problems

What is Linear Programming?

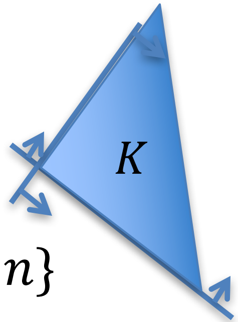
- Everyone knows how to write down a linear program

$$\min c^\top x$$
$$\text{s.t. } Ax = b$$
$$x \geq 0$$

- Nonnegativity constraint is what makes standard form LP meaningful
- In general, linear inequality makes LP meaningful
  - $Ax \geq b$ is equivalent to $Ax - b = s,\ s \geq 0$
- $x \geq 0$ is equivalent to say $x \in R_+^n$, where $R_+^n \coloneqq \{x \in R^n : x_i \geq 0\ \forall i = 1, \ldots, n\}$ is the nonnegative orthant

What is the Nonnegative Orthant?

- We have learned convex cones:
  - A set K is called a **convex cone** if K is convex and $\alpha x \in K$ for all $\alpha \geq 0$ whenever $x \in K$



- Nonnegative orthant $R_+^n = \{x \in R^n \mid x_i \geq 0, \forall i = 1, \dots n\}$ is a convex cone
  - Because if $x \in R_+^n$, then $\alpha x = (\alpha x_1, \dots, \alpha x_n) \in R_+^n$ for any $\alpha \geq 0$
  - And R_+^n is a polyhedron defined by halfspaces $x_i \geq 0$, thus convex

What is Relation between Order and Cone?

- Order is a comparison relationship between two elements a and b, usually written as $a \geq b$ (when they are comparable)
- For example, when a, b are two vectors, we say $a \geq b$ if
  - $a - b \geq 0$, i.e. $a_i - b_i \geq 0 \; \forall i$
  - Equivalent to $a - b \in R_+^n$
  - So $a \geq b$ is the same as saying $a - b \in R_+^n$
  - $\geq$ can be viewed as being defined by the nonnegative orthant cone
  - To emphasize, we write $a \geq_{R_+^n} b$
- What is the relation of an order and a convex cone?
  - An order $\succcurlyeq_K$ is defined by an underlying convex cone K as
    - $a \succcurlyeq_K b$ if and only if $a - b \in K$

How to Generalize Linear Programming?

- A standard form LP can be viewed as

$$\min\{c^\top x : \ Ax = b, x \geq_{R_+^n} 0\}$$

- An **elegant** way to generalize linear programming is to generalize R_+^n to a general convex cone K

  Linear Conic Programming: $\min\{c^\top x : \ Ax = b, x \geq_K 0\}$

- **Generalized inequality** defined by convex cone K: $Ax \geq_K b$
  - This inequality is no longer linear!

Summary

- We learned:
  - How to understand linear programming from a conic perspective
  - How to generalize linear programming to linear conic programming